Synthetic Data Design Task

Design Task: A Synthetic Data Engine for Next-Generation Temporal AI

Generating, validating, and using synthetic data at scale

About this task. This task combines system design with an empirical prototype. We want to understand how you structure the problem, choose among competing approaches, and reason about trade-offs. We do not expect a production-ready implementation. The proof of concept should support one credible conclusion: a small, well-controlled experiment is more valuable than a large number of inconclusive runs.

Our ambition

Aionic Labs builds temporal AI: models that understand systems as they evolve through time, from a machine or a patient to a power grid. We believe the field of Temporal Foundation Models and Temporal AI is still in its infancy. While Time-Series Foundation Models already exist, they have not yet reached the level of generality we associate with foundation models in language and vision. Models such as ChatGPT can operate across a broad range of tasks and modalities with a common underlying model, whereas today's time-series models remain comparatively constrained across domains, tasks, modalities, and temporal scales.

Our goal is to close that gap and build increasingly general-purpose models for understanding and reasoning about systems as they evolve over time. One important step in this direction is to develop models that can perform multiple time-series tasks within a single model, conditioned on natural-language instructions and context, for example, forecasting future trajectories, classifying system states, detecting and explaining events, answering questions about a signal, imputing missing observations, editing or generating signals, and reasoning about possible interventions.

Getting enough training data is a major bottleneck. Raw sensor data is abundant across healthcare, wearables, industrial systems, energy, and finance, but high-quality language supervision is scarce, expensive, and unevenly distributed. Human annotation alone cannot provide the scale and coverage required for foundation-model training. Synthetic data is a potential technical moat for Aionic.

What we mean by an engine

We want to see how you turn this capability into a reliable data engine. A engine should identify valuable gaps in the corpus, generate targeted examples, evaluate their quality, reject unsuitable outputs, record complete provenance, and measure whether the retained data improves model performance. Generation is one part of this system.

We expect you to define the components and boundaries you consider necessary. The following questions illustrate the scope:

- How should the system decide what to generate, rather than simply reproduce the existing data distribution?
- Which generation methods are appropriate, and when might the system need more than one?
- How can it assess plausibility, diversity, semantic alignment, and memorization at scale?
- What provenance, validation results, and version information should each synthetic example retain?
- Where does expert review provide the greatest value?
- How should synthetic-data quality connect to downstream model performance?

The task and proof of concept

Design a synthetic data engine for a model of your choice, ideally one you know well. Outline how it generates examples, checks quality and diversity, avoids memorization, and integrates synthetic data into training. Approaches could include semantic-to-signal generation, CLIP/unCLIP-style latent decoding, controlled signal editing, simulation, or another defensible approach.

Build a small proof of concept focused on synthetic data generation. Choose a public dataset, generate synthetic examples, and compare training with and without them on an untouched real-data holdout set. Explain the results, whether positive or negative, and what you learned. You do not need to implement the broader engine.

> **Central question.** *Can your synthetic data improve model performance on held-out real data?*

A well-supported negative result is as valuable as a positive one. We care about an evaluation we can trust: use a defensible baseline, isolate the effect of synthetic data, prevent leakage between generation, training, and

evaluation, and control for differences in training budget. Where feasible, repeat runs to show that the observed effect exceeds normal variation. Explain what evidence would change your conclusion and what you would run next with one additional week and a larger cluster.

Practicalities

Please spend no more than three days on this task. Prepare a short presentation of your design and experiment for a 45-minute discussion. Anything you can share beforehand, such as slides, notes, or code, is appreciated so we can review it.

References

OpenTSLM repository

TimeNet documentation